

3 Increased use of wireless devices at the workplace

Tablet computers and smartphones have become so popular that they have a significant presence at the workplace too—not as devices under test but as part of the test system. While the computing power made available by these devices is notable, they cannot replace the PC and related measurement platforms like

PXI. Instead, these devices are suitable for data consumption and report viewing, and system monitoring and control.

National Instruments' Automated Test Outlook 2012 explains: "The explosion of mobile devices like tablets and smartphones provides compelling benefits to engineers, technicians

and managers involved in automated test who need remote access to test status information and results. While today's technology offers solutions for monitoring or remote reporting via mobile devices, test organisations will need new expertise to unite the networking, Web services and mobile app portions of the solution."

4 Software-defined instrumentation

As the complexity of products continues to increase, their testing becomes much more challenging. Test engineers now require test systems that are flexible enough to support the wide variety of tests that must be performed on a single product while being scalable enough to encompass a larger number of tests as new functionality continues to be added.

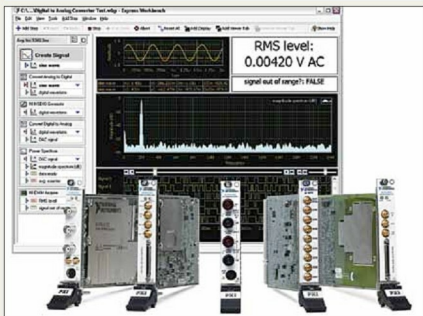
"Increasingly, the functionality of complex devices is being defined by the software embedded in them. This is challenging for many test engineers because most standalone instruments cannot change their functionality as fast as changes in the device under test (DUT) due to the fixed user interface and firmware that must be developed and embedded in the instrument. Thus test engineers are turning to a software-defined approach to instrumentation, so that they can quickly customise their equipment to meet specific application needs and integrate testing directly into the design process," says Eric Starkloff, director of NI Test Product Marketing.

Thakare shares two major advantages of software-defined instrumentation: "First, it can dramatically reduce the number of hardware components in all the mixed-signal designs, which means smaller chip size for system-on-chip implementation. Second, it can provide automatic adjustment or compensation for circuit component variations due to temperature dependence, ageing and manufacturing tolerances."

Test engineering: A strategic asset

According to NI's recent global survey of test engineering leaders, a shift has emerged within electronics manufacturing firms wherein they are now using product test for competitive differentiation. This shift culminates in an elevation of the test engineering function to become a strategic asset to the company. The test participants of the survey said that their primary goal over the next few years was to reorganise their test organisation structures for enhanced efficiency.

(Source: NI Automated Test Outlook 2012)



Software-defined instrumentation by NI (Image courtesy: www.ni.com)

Software-defined instrumentation looks to become an essential component of scalable and highly performing test systems. Singh agrees by saying, "We predict a bright future for software-defined instrumentation. Software-defined instruments, also known as virtual instruments, are modular hardware with user-defined software giv-

ing the flexibility to combine standard and user-defined measurements with custom data processing using common hardware components. This flexibility is useful for electronic devices like advanced navigation systems and communication devices like smartphones to integrate diverse capabilities and adopt new communication standards."